

Latent Rectified Flow for Medical Image Reconstruction

CS 231N Milestone 2: Technical Approach

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Approach Shift & Intuition

- **The Baseline Limitation:** Standard U-Nets optimize for pixel averages (MSE), producing blurry, "safe" images that lose critical medical textures.
- **The Generative Shift:** We are shifting to Rectified Flow. Because we have perfectly paired data (Low-Dose and Normal-Dose CTs), we don't need to diffuse to pure Gaussian noise like standard DDPMs.
- **The Key Insight:** We can learn a straight-line vector field directly between the two domains, transporting Low-Dose to High-Dose along the shortest path.
- **Why It Matters:** A straight path means fewer ODE steps at inference — closer to real-time clinical use.

Method Overview: The Full Pipeline

- **Step 1 – Compression:** Pass the noisy 512x512 Low-Dose CT slice through a pre-trained, frozen VAE encoder to obtain a compact latent representation (Z_0).
- **Step 2 – Rectified Flow:** Pass Z_0 through our trained Transformer/U-Net vector-field model to transport it along a straight path to the clean latent (Z_1).
- **Step 3 – Decoding:** Pass the generated Z_1 through the VAE decoder to recover the high-fidelity pixel-space image.
- **Why Latent + Flow:** Latents shrink dimensionality ~90%; straight flows shrink the number of solver steps. Together: feasible on a single GPU.

Design Choice 2 — Paired Rectified Flow Math

- **The "Why":** Standard diffusion requires hundreds of steps to reverse complex noise curves. Rectified Flow learns an ODE that draws the shortest straight path, decoding in just 5–10 steps.
- **The Interpolation:** We define a linear interpolation between paired latents: $Z_t = t \cdot Z_1 + (1 - t) \cdot Z_0$.
- **The Objective:** Our network $v_\theta(Z_t, t)$ is trained to predict the velocity (vector derivative) pointing from Z_0 to Z_1 .
- **The Payoff:** Fewer function evaluations at inference and a well-defined, invertible ODE — properties we will exploit for uncertainty estimation.

Loss Functions & Training Setup

- **Velocity Matching Loss:** Primary objective of the Rectified Flow network:
$$L_{\text{flow}} = \|v_{\theta}(Z_t, t) - (Z_1 - Z_0)\|^2.$$
- **Latent Pairing:** For each training step, sample a paired (Low-Dose, Normal-Dose) slice, encode both, sample $t \in [0, 1]$, and form the interpolated latent Z_t .
- **Optimizer & Schedule:** AdamW with cosine LR decay; mixed-precision training on a single GPU.
- **Compute Budget:** Target end-to-end training and evaluation under the \$100 cap by exploiting the latent compression and small step count.

Experimental Setup & Baselines

- **Dataset:** Mayo Clinic Low-Dose CT Grand Challenge. We will slice the 3D volumes into 2D axial slices for training efficiency.
- **Baseline:** A classic Residual U-Net performing direct image-to-image mapping — no latent space, no flow.
- **Structural Metric:** SSIM (Structural Similarity Index) to confirm organ boundaries are preserved.
- **Generative Metric:** FID (Fréchet Inception Distance) to demonstrate our flow model produces more realistic textures than the U-Net baseline.

The Clinical Edge – Probability & Uncertainty

- **The Problem:** Standard ODE solvers used in Rectified Flow are perfectly deterministic. The same input yields the same output every time – radiologists have no way to know when the model is "guessing."
- **Our Solution:** Introduce a probabilistic sampling layer to evaluate per-pixel confidence alongside the reconstruction.
- **Two Paths Forward:** We are exploring two mathematically distinct approaches – one based on stochastic sampling, one on exact likelihood – and will compare which is more useful clinically.
- **The Output:** A reconstructed image plus a confidence map flagging regions where the model may have hallucinated detail.

Uncertainty Paths – Option A vs. Option B

- **Option A – SDE Variance (Monte Carlo proxy):** Switch the solver from an ODE to an SDE, injecting Langevin noise at inference. Run the same slice 10 times to generate 10 slightly different clean images.
 - Output: a "Variance Heatmap" – high pixel variance indicates low confidence and possible hallucination.
- **Option B – Exact Log-Likelihood (Continuous Normalizing Flows):** Because Rectified Flow is an ODE, it acts as a Continuous Normalizing Flow. Using the Hutchinson trace estimator, we can compute the exact change in probability density from Z_0 to Z_1 .
 - Output: a formal $\log p(x)$ score – an exact mathematical likelihood that the generated image belongs to the distribution of true medical scans.